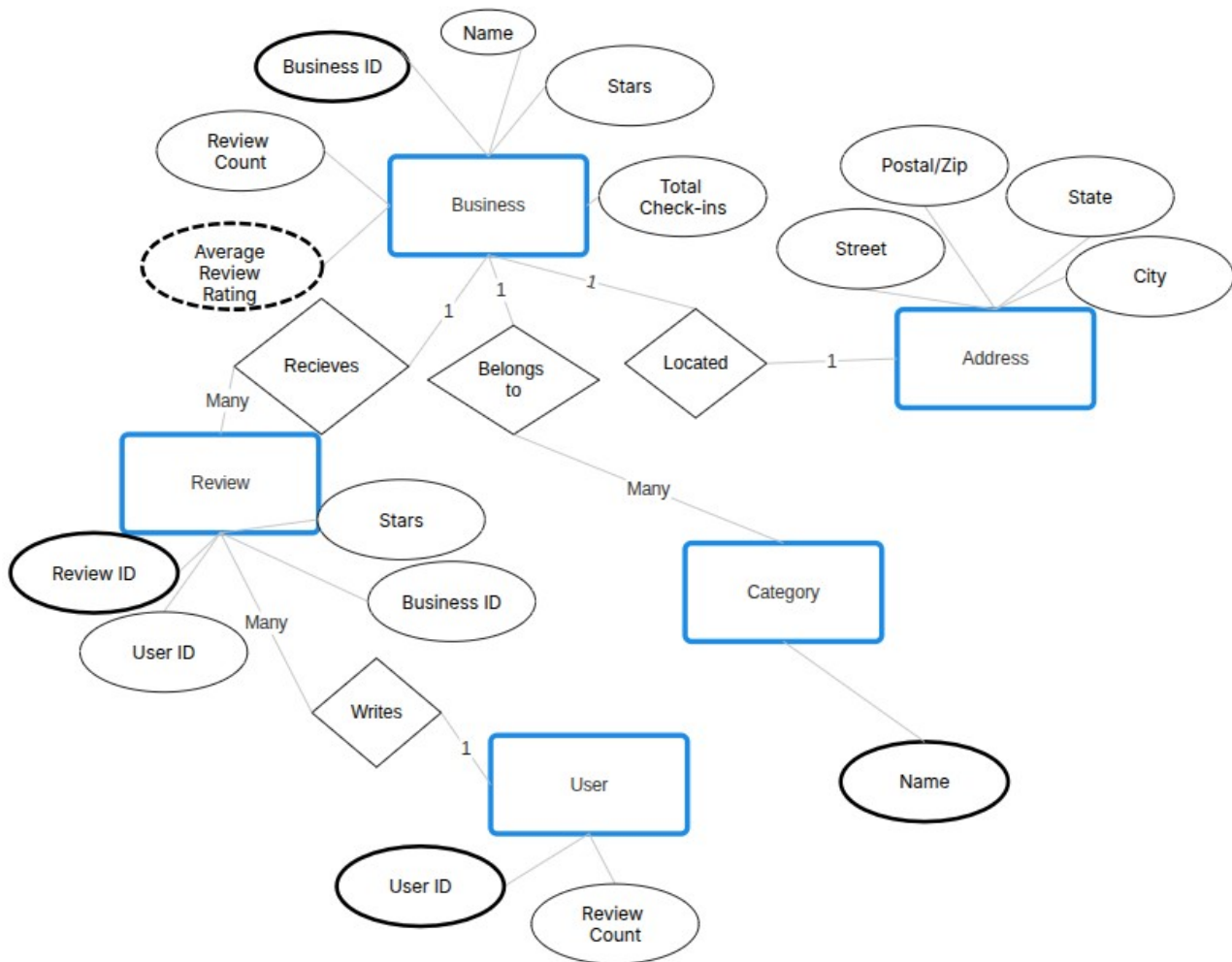


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2i) Design a database schema that models the database for the described application scenario in Appendix-A and provide the ER diagram for your database design:



2 ii) Translate the ER diagram into relations with DDL SQL statements for creating tables in a relational DBMS:

Entities

Business:

1. Business ID (pk)
2. Name
3. Stars
4. Total Check-ins
5. Review Count
6. Average Review Rating (fk, references Review)

Review:

1. Review ID (pk)
2. User ID (fk, references User)
3. Business ID (fk, references Business)

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4. Stars

Address:

1. Street
2. Postal/Zipcode
3. State
4. City

Category:

1. Category Name (pk)

User:

1. User ID (pk)
2. Review Count

Relations

Located: Connects business to a composite address entity, allows easier querying of individual components (like zip) One to one, business ID will serve as the primary key for the address.

Receives: Connects business with review. One business may have many reviews. User ID and Business ID are foreign keys, reliant on User and Business entities.

Belongs to: Connects business with category. One business may be included in many categories, and categories can include many businesses.

Writes: Connects review to user. One user may write many reviews.

SQL Code: (boothby_relations_v1.sql in github repo)

```
-- For part 2 of Milestone 1, translate ER model into SQL DDL statements
```

```
--to reset db
DROP TABLE IF EXISTS Business_Category;
DROP TABLE IF EXISTS Review;
DROP TABLE IF EXISTS Business;
DROP TABLE IF EXISTS Category;
DROP TABLE IF EXISTS "user";
DROP TABLE IF EXISTS Address;
```

```
create table Address(
addressID SERIAL PRIMARY KEY, --creates a unique id to use as primary key
street VARCHAR(255) NOT NULL,
zipcode VARCHAR(5) NOT NULL,
city VARCHAR(50) NOT NULL,
state VARCHAR(2) NOT NULL -- US state abbrev.
);
```

```
create table Business (
```

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```
businessID INT PRIMARY KEY,  
name varchar(255) not null,  
stars NUMERIC(2,1) NOT NULL,  
totalCheckins INT,  
reviewCount INT DEFAULT 0,  
averageReviewRating NUMERIC(2,1), --calculated from each review
```

```
--components inherited from Address  
addressID INT NOT NULL,
```

```
FOREIGN KEY (addressID) REFERENCES Address(addressID)  
);
```

```
create table "user" (  
userID INT PRIMARY KEY,  
reviewCount INT DEFAULT 0  
);
```

```
create table Review (  
reviewID INT PRIMARY KEY,  
userID INT NOT NULL,  
stars NUMERIC(2,1) NOT NULL,  
businessID INT NOT NULL,
```

```
FOREIGN KEY (businessID) REFERENCES Business(businessID),  
FOREIGN KEY (userID) REFERENCES "user"(userID)  
);
```

```
create table Category(  
categoryTitle VARCHAR(255) PRIMARY KEY  
);
```

```
create table Business_Category (  
businessID INT NOT NULL,  
categoryTitle VARCHAR(255) NOT NULL,  
PRIMARY KEY (businessID, categoryTitle),  
FOREIGN KEY (businessID) REFERENCES Business(businessID),  
FOREIGN KEY (categoryTitle) REFERENCES Category(categoryTitle)  
);
```

3. Create simple postgresSQL database adapting milestone1.csv: (milestone1DB.sql in github repo)

```
--For part 3 of Milestone 1, references "milestone1DB.csv"
```

```
create table Business(  

```

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```
name VARCHAR(255) PRIMARY KEY,  
state VARCHAR(10) NOT NULL, --no fields can be null, states are always 2 letter abbrev.  
city VARCHAR(50) NOT NULL  
);
```

3ii. Write a python app to connect milestone1DB and run queries on the business table (milestone1.py in github repo)

```
#skeleton code from professor  
import psycopg2  
import sys  
from PyQt6.QtWidgets import QMainWindow, QApplication, QWidget,  
QTableWidget, QTableWidgetItem, QVBoxLayout  
from PyQt6 import uic, QtCore  
from PyQt6.QtGui import QIcon, QPixmap
```

```
qtCreatorFile = "milestone1App.ui"  
Ui_MainWindow, QtBaseClass = uic.loadUiType(qtCreatorFile)
```

```
class milestone1(QMainWindow):  
    def __init__(self):  
        super(milestone1, self).__init__()  
        self.ui = Ui_MainWindow()  
        self.ui.setupUi(self)  
        try:  
            self.conn = psycopg2.connect(  
                database="milestone1db",  
                user="sydnee"  
            )  
            self.cursor = self.conn.cursor()  
            print("Database connected!")  
        except Exception as e:  
            print(f"Database connection error: {e}")  
        return  
        self.ui.stateComboBox.currentIndexChanged.connect(self.on_state_selected)  
        self.ui.cityComboBox.currentTextChanged.connect(self.on_city_selected)  
        self.load_states()
```

```
#load states from csv  
def load_states(self):  
    try:  
        self.cursor.execute("""  
            SELECT DISTINCT state  
            FROM business  
            ORDER BY state  
            """)  
        states = self.cursor.fetchall()  
        print(f"Found {len(states)} states")
```

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```
self.ui.stateComboBox.clear()
self.ui.stateComboBox.addItem("") # Empty option
for state in states:
    self.ui.stateComboBox.addItem(state[0])
    print(f"Added state: {state[0]}")
except Exception as e:
    print(f"Error loading states: {e}")

# print to console check
print(self.ui.stateComboBox)
print(self.ui.cityComboBox)
print(self.ui.businessTable)
def on_state_selected(self):
    selected_state = self.ui.stateComboBox.currentText()
    print(f"State selected: {selected_state}")
    # display cities
    if not selected_state:
        self.ui.cityComboBox.clear()
        self.ui.businessTable.setRowCount(0)
        return
    try: # query to fetch all cities for given state
        self.cursor.execute("""
        SELECT DISTINCT city
        FROM business
        WHERE state = %s
        ORDER BY city
        """, (selected_state,))
        cities = self.cursor.fetchall()
        print(f"Found {len(cities)} in state {selected_state}")
        self.ui.cityComboBox.clear()
        self.ui.cityComboBox.addItem("")

    for city in cities: # print all cities in ui
        self.ui.cityComboBox.addItem(city[0])
        print(f"Added city: {city[0]}")
        self.ui.businessTable.setRowCount(0)

except Exception as e:
    print(f"Error loading cities for state {selected_state}: {e}")

def on_city_selected(self):
    state = self.ui.stateComboBox.currentText()
    item = self.ui.cityComboBox.currentItem()

    if not item:
        self.ui.businessTable.setRowCount(0)
        return
    city = item.text()
```

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```
# # DEBUG
# print(f"DEBUG: state = '{state}'")
# print(f"DEBUG: city = '{city}'")

if not state or not city:
self.ui.businessTable.setRowCount(0)
return
try: #query to fetch all businesses
self.cursor.execute("""
SELECT name, city, state
FROM business
WHERE city = %s and state = %s
ORDER BY name
""", (city, state))
businesses = self.cursor.fetchall()

# print(f"Found {len(businesses)} businesses in {city}, {state}")
# print(f"DEBUG: businesses data = {businesses}")

self.ui.businessTable.setColumnCount(3) #columns for name, city, and state
self.ui.businessTable.setRowCount(len(businesses))

for row_idx, business in enumerate(businesses):
for col_idx, value in enumerate(business):
self.ui.businessTable.setItem(row_idx, col_idx, QTableWidgetItem(str(value)))

except Exception as e:
print(f"Error loading businesses: {e}")

if __name__ == "__main__":
app = QApplication(sys.argv)
window = milestone1()
window.show()
sys.exit(app.exec())
```

Demonstration Image:

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The screenshot shows a PyCharm IDE with a project named "ProjectMilestone1". The Explorer panel on the left shows the project structure, including files like `milestone1DB.sql`, `milestone1DB.csv`, `milestone1.py`, and `milestone1App.ui`. The main editor displays the `milestone1.py` file, which contains the following code:

```
1  ## Project Milestone 1
6
LEGES ON SCHEMA public TO username;"

/milestone1DB.csv' DELIMITER ',' CSV QUOTE ''
```

The application window, titled "MainWindow", is divided into two panes. The top pane, "Milestone1-Page1", contains a form for "State / City" and a table of businesses. The "State" dropdown is set to "AZ", and the "City" dropdown is set to "Avondale". The table lists businesses in Avondale, AZ, with the following data:

	1	2	3
1	1 Brothers ...	Avondale	AZ
2	A & D ...	Avondale	AZ
3	Aesthetic ...	Avondale	AZ
4	Amazon ...	Avondale	AZ
5	Ann Nails	Avondale	AZ
6	Avondale ...	Avondale	AZ
7	Avondale ...	Avondale	AZ
8	Avondale ...	Avondale	AZ
9	A&W	Avondale	AZ

The bottom pane, "Milestone2-Page2", displays a list of businesses. The terminal at the bottom shows the command to run the application:

```
(. venv) sydney@syds-laptop:~/cpts451/ProjectMilestone1$ python milestone1.py
```

The terminal output shows a list of businesses in Avondale, AZ, including:

- 's Mexican Restaurant', 'Avondale', 'AZ'), ('Mariscos Las Playitas', 'Avondale', 'AZ'), ('McFall Tire and Auto Repair', 'Avondale', 'AZ'), ('Mirabella Luxury Apartment Homes', 'Avondale', 'AZ'), ('MVP Kids Care', 'Avondale', 'AZ'), ('My Nails Boutique and Spa', 'Avondale', 'AZ'), ('New Wok Chinese Food', 'Avondale', 'AZ'), ('N-T Nails & Spa', 'Avondale', 'AZ'), ('Phade City Barbershop', 'Avondale', 'AZ'), ('Phoenix International Raceway', 'Avondale', 'AZ'), ('Pilot Travel Center # 459', 'Avondale', 'AZ'), ('Raul and Theresa's Restaurant', 'Avondale', 'AZ'), ('Rio Santa Fe', 'Avondale', 'AZ'), ('Screwballs Pizza & Spirits', 'Avondale', 'AZ'), ('Starlight Cuisine', 'Avondale', 'AZ'), ('St Thomas Aquinas Church', 'Avondale', 'AZ'), ('Sunny Internal Medicine and Pediatrics', 'Avondale', 'AZ'), ('Thai Orchid Garden Restaurant', 'Avondale', 'AZ'), ('The Sanctuary Nails and Spa', 'Avondale', 'AZ'), ('Tony's Cafe', 'Avondale', 'AZ'), ('T Pro Nails & Spa', 'Avondale', 'AZ'), ('Ubaldo's Taco Shop', 'Avondale', 'AZ'), ('Universal Technical Institute', 'Avondale', 'AZ'), ('Valley Golf Center', 'Avondale', 'AZ'), ('VS Nails & Spa', 'Avondale', 'AZ'), ('Walt's BBQ', 'Avondale', 'AZ'), ('Waterworks Water & Ice', 'Avondale', 'AZ'), ('Weiss Guys Self Service Car Wash', 'Avondale', 'AZ'), ('Western Dry Cleaners', 'Avondale', 'AZ'), ('Western Laundry Mats', 'Avondale', 'AZ'), ('West Valley Pediatrics', 'Avondale', 'AZ'), ('Win Nails', 'Avondale', 'AZ'), ('Xpress Care Pharmacy', 'Avondale', 'AZ')]